

Operating System & System Programming (OSSP)

Lab File: Linux File Permissions & File Handling in C

1. C Source Code: permission.c

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    FILE *fp;
    fp = fopen("sample.txt", "w");
    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 1;
    }
    fprintf(fp, "Hello, this is a test file.\n");
    fclose(fp);
    printf("File created successfully.\n");
    return 0;
}
```

2. Initial Compilation & Execution

```
rohit_siwatch@RohitSiwatch:~$ nano permission.c
rohit_siwatch@RohitSiwatch:~$ gcc permission.c -o permission
rohit_siwatch@RohitSiwatch:~$ ./permission
File created successfully.
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
-rw-r--r-- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
rohit_siwatch@RohitSiwatch:~$ chmod 666 sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
-rw-rw-rw- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
rohit_siwatch@RohitSiwatch:~$ chmod u+rwx sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
-rw-rw-rw- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
```

3. File Permission Tasks

Task 1: Give Read and Write only to the owner

```
rohit_siwatch@RohitSiwatch:~$ chmod 600 sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
```

```
-rw----- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
```

Task 2: Give Read and Write to the owner and group

```
rohit_siwatch@RohitSiwatch:~$ chmod 660 sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
-rw-rw---- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
```

Task 3: Only owner can read and write

```
rohit_siwatch@RohitSiwatch:~$ chmod 600 sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
-rw----- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
```

Task 4: Owner read/write, everyone else read-only

```
rohit_siwatch@RohitSiwatch:~$ chmod 644 sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
-rw-r--r-- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
```

Task 5: Owner can do everything, others can only read and execute

```
rohit_siwatch@RohitSiwatch:~$ chmod 755 sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
-rwxr-xr-x 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
```

Task 6: Owner and group can read/write

```
rohit_siwatch@RohitSiwatch:~$ chmod 660 sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
-rw-rw---- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
```

Task 7: Only owner can execute

```
rohit_siwatch@RohitSiwatch:~$ chmod 100 sample.txt
rohit_siwatch@RohitSiwatch:~$ ls -l sample.txt
---x----- 1 rohit_siwatch rohit_siwatch 28 Aug 17 05:22 sample.txt
```

4. Summary of Permission Rules

Rule No.	Permissions Specification	Octal Mode
Rule 8	Owner: read + write; Group: nothing; Others: nothing	600
Rule 9	Owner: read + write; Group: read; Others: read	644
Rule 10	Owner: read + write + execute; Group: read + execute; Others: read + execute	755
Rule 11	Owner: read + write; Group: read + write; Others: nothing	660
Rule 12	Everyone: read only	444